

Enhancing Machine Learning Classification of Cadmium and Lead Using DWT Adaptive filter Based Artifact Reduction

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Abstract— Motion artifacts caused by electrode movement are a significant source of signal interference in voltammetry measurements, which can reduce the performance of machine learning classification models. This study proposes and evaluates an approach to improve the classification accuracy of voltammetry signals for Cadmium, Lead, and Non-Heavy Metal categories through noise reduction. The proposed method used is the Discrete Wavelet Transform (DWT) Adaptive Filter 6 level decomposite to clean the signal before statistical feature extraction. The filtered signal is then used to train four classification models: Support Vector Machine (SVM), K-Nearest Neighbors (KNN), XGBoost, and Random Forest. Evaluation results show that the application of the DWT Adaptive Filter dramatically improves the performance of all models. The KNN model achieved the highest F1-Score of 0.97, followed by XGBoost with an F1-Score of 0.96, far surpassing the best performance under baseline conditions (without filtering), which only reached 0.87. This substantial improvement demonstrates that the DWT-based filtering approach is highly effective in enhancing signal quality, which directly results in significantly more accurate and reliable classification.

Keywords—Adaptive Filtering, Classification, DWT, Machine Learning, Motion Artifact, and Voltammetry.

I. INTRODUCTION

Motion artifact is a signal disturbance that arises due to the movement of electrodes or sensors during the measurement process on biomedical wearable devices [1]. This motion causes changes in the contact between the electrode and the skin or sample surface resulting in a signal that does not reflect the actual physiological or chemical conditions [2]. An instance is the application of Photoplethysmography (PPG) sensors, that analyse changes in blood volume through light reflection. The sensor's position is altered by hand or body movement, resulting in the generation of noise. In voltammetry sensors, shifting of the electrodes against the solution or sample surface during

measurement can give rise to unwanted signal fluctuations. Similarly, in e-nose (electronic nose) sensors that use semiconductor-based gas sensors, movement of the sensor or sample can cause transient changes in local gas concentration, resulting in noise in the recorded data. The primary causative mechanisms of motion artifacts in voltammetric measurements comprise abrupt changes in the double-layer capacitance at the electrode-electrolyte interface and disturbances in analyte mass transport [1], [2]. These disturbances can mask the target signal, leading to quantification errors or the generation of false positive results, particularly in real-time monitoring applications where continuous measurements are required. Addressing this challenge necessitates robust signal processing strategies that can adaptively respond to varying noise characteristics while preserving essential electrochemical features.

The performance of classification and prediction models in electrochemical signal processing is greatly affected by noise caused by motion artifacts and other environmental disturbances. Noisy data can hinder the efficacy of Machine Learning (ML) and Deep Learning (DL) models in recognizing underlying patterns, potentially leading to decreased model accuracy, increased misclassification rates, and reduced model stability, particularly in complex time-series data such as voltammetric signals [4], [5]. Employing appropriate filtering strategies is essential to improve the quality of the input signal, thereby ensuring optimal performance of classification models. Various studies have demonstrated that employing noise reduction techniques, including DWT, Kalman filtering, and clustering methods, can significantly improve the accuracy and performance of classification and prediction models [6], [7]. When adequate filtering is in place, ML models including SVM, KNN, XGBoost, and Random Forest can achieve more accurate classification results and more reliable predictions.

A variety of filtering methods have been employed in the literature to address noise in biomedical signals and electrochemical sensors. These include DWT, which is capable of denoising while maintaining signal information [8]; Kalman Filtering, which uses an adaptive prediction model [9]; Savitzky-Golay filters, which preserve peak shapes [10]; and Moving Average Filters, which are simple and easy to implement [11]. Furthermore, unsupervised learning methods such as K-means clustering are employed to group data and detect centroids of clean signals [6]. However, each method has inherent limitations when dealing with complex noise and dynamic fluctuations in real-world signals. Traditional DWT approaches often rely on fixed thresholding parameters that may not optimally adapt to varying noise levels across different measurements.

While existing studies have applied wavelet-based denoising to various biosignals, limited research has focused on developing adaptive filtering specifically tailored for voltammetric signals affected by motion artifacts. Most approaches use fixed wavelet parameters and generic thresholding strategies that do not account for the unique characteristics of electrochemical signals. Our contribution lies in developing an adaptive DWT filtering framework where the threshold values are dynamically calculated based on actual noise levels estimated from the signal itself, combined with systematic outlier detection using PCA and K-means clustering. This integrated approach provides a more robust solution for real-world electrochemical sensor applications.

The present study proposes an adaptive filtering approach combining Principal Component Analysis (PCA) and K-means clustering for outlier detection, followed by adaptive DWT filtering with dynamically optimized parameters. The adaptiveness is achieved through: (1) automatic noise level estimation using the median absolute deviation of first-level detail coefficients, (2) dynamic threshold calculation based on signal length and estimated noise variance, and (3) level-specific thresholding that accounts for varying noise distributions across decomposition levels. The filtered signal results are evaluated using four machine learning classification models (SVM, KNN, XGBoost, and Random Forest) through rigorous cross-validation protocols. This approach is expected to improve the accuracy and reliability of classifying voltammetric signals affected by motion artifacts, with direct applicability to environmental monitoring and biomedical sensing applications.

This work presents a significant contribution in the form of the development of an adaptive filtering framework using Discrete Wavelet Transform (DWT), specifically designed for voltammetric signals disturbed by motion artifacts. The fundamental innovation of this approach lies in the integration of adaptive filtering with unsupervised outlier detection, leveraging a combination of Principal Component Analysis (PCA) and K-means. The performance advantages of the proposed framework have been statistically validated through comprehensive comparative analysis with various standard filtering methods and rigorous cross-validation testing. Furthermore, the findings of this study substantiate the practical implementation and efficacy of the proposed

method in authentic environmental monitoring scenarios, particularly in the context of heavy metal detection.

II. LITERATURE REVIEW

A. Voltammetric Data

An electrochemical method known as Cyclic Voltammetry (CV) is often employed to examine intricate complexation processes, particularly in the field of environmental chemistry to gain insight into the binding of heavy metals to organic ligands [3], [4], [5]. Voltammograms revealed the relationship between current and potential, providing insights into the electrochemical reactions within the sample. This was done using a glassy carbon electrode (GCE) as the working electrode and a sintered reference electrode (RE) for potential stability. [6].

B. Motion Artifact

Motion artifacts in voltammetric signals are significant non-chemical disturbances caused by the physical movement of electrodes, posing a major challenge, especially in in vivo measurements or in wearable sensors [1], [2]. The primary causative mechanisms comprise abrupt changes in the double-layer capacitance at the electrode-electrolyte interface and disturbances in analyte mass transport. These disturbances manifest as sharp current spikes or baseline drifts in the data, with amplitudes frequently exceeding the amplitude of the analyte signal itself (signal-to-artifact ratios can reach 1:10 or worse) [3]. This can result in masking of the target signal, leading to quantification errors or generation of false positive results. Unlike thermal noise or electronic interference, motion artifacts are characterized by their non-stationary nature and unpredictable occurrence patterns, making them particularly challenging to filter using conventional frequency-domain methods.

Recent studies have shown that motion artifacts in electrochemical sensors share similarities with those in other biomedical signals but exhibit unique characteristics related to electrochemical kinetics [1]. The time scale of motion artifacts (typically 0.1-10 Hz) often overlaps with the relevant electrochemical signal frequencies, making simple band-pass filtering inadequate. This overlap necessitates more sophisticated signal processing approaches that can discriminate between artifact-induced fluctuations and genuine electrochemical responses.

C. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction method that simplifies complex data by converting it to a novel coordinate system called Principal Components (PC) [7]. The ordering of these axes is based on their capacity to capture the maximum variation in the data, thereby ensuring that the initial components can represent the majority of the salient information.

Its primary function in outlier detection is its capacity to circumvent the "Curse of Dimensionality" in high-dimensional data, thereby elucidating anomalies [8]. By concentrating the analysis on the predominant patterns of variation, normal data will have a tendency to congregate together, while outliers that deviate from these patterns will be projected away from the primary group. This separation

enables the effective identification of outliers through both computational and visual means. The equation of data standardization in PCA in Eq. (1) :

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \quad (1)$$

The subsequent Eq. (1) by transforming each original data value (x_{ij}) into a standardized value (z_{ij}). This method entails calculating the disparity between the initial value and the mean (μ_j) of the relevant feature, and then dividing that disparity by the standard deviation (σ_j) of the same feature. This transformation seeks to normalize each feature to achieve a mean of 0 and a standard deviation of 1.

D. K-Means Clustering

K-Means is a clustering technique designed to divide n observations into K specified clusters. subsequently is suggested that each observation will belong to the cluster with the nearest average, or centroid, as established by the measurement of central tendency [9].

The K-means approach enhances clustering by minimizing the Within-Cluster Sum of Squares (WCSS), defined as the sum of the squared Euclidean distances between data points and the corresponding centroids. Eq. (2) demonstrates how this objective function operates.

$$J = \sum_{k=1}^K \sum_{x_i \in s_k} \|x_i - \mu_k\|^2 \quad (2)$$

The subsequent Eq. (2) discussion will elucidate the mechanics of this formula. For each cluster (from $k = 1$ to K), the algorithm first calculates the squared Euclidean distance ($\|x_i - \mu_k\|^2$) relationship every single data point (x_i) and the center point (μ_k) in the respective cluster. Subsequently, the squared distance values within each cluster are aggregated to calculate the total distance. Ultimately, the aggregate of all these per-cluster sums is aggregated to obtain a final value of J . This value of J signifies the total internal variation of all clusters, with smaller values denoting superior and more compact clustering.

E. Discrete Wavelet Transform (DWT)

The signal filtering process employing the discrete wavelet transform (DWT) is a systematic procedure comprising three primary stages that are designed to remove noise [10]. The process begins with the decomposition of the initial signal, which is subsequently broken down into two distinct types of components in a multi-level approach [11]. The coefficient of approximation (cA) is the primary metric responsible for storing the core information or the main trend of the signal. In contrast, the coefficient of detail (cD) is responsible for capturing high-frequency information, which is typically where noise resides. The decomposition value is defined as Eq. (3) and Eq. (4):

$$cA_{j,k} = \sum_n x[n] \cdot G_{j,k}[n] \quad (3)$$

$$cD_{j,k} = \sum_n x[n] \cdot H_{j,k}[n] \quad (4)$$

In the DWT decomposition process Eq. (3) and Eq. (4), the original input signal, denoted as $x[n]$, is decomposed into its basic components. This process is achieved by passing the signal through a pair of complementary filters: a low-pass filter, represented as $G[n]$, and a high-pass filter, represented as $H[n]$. The output of the low-pass filter $G[n]$ is the approximation coefficient ($cA_{j,k}$), which represents the main trend or low-frequency component of the signal at decomposition level j . Conversely, the output of the high-pass filter is the approximation coefficient ($cA_{j,k}$). Conversely, The result produced by the high-pass filter $H[n]$, is the Detail Coefficient ($cD_{j,k}$), which captures high-frequency information, including details, edges, and noise, at the same level j . Consequently, this decomposition effectively separates the essence of the signal from its details.

The next crucial step is thresholding, a process in which a threshold value (T) is selectively applied to only the detail coefficients to suppress or remove components that are considered as noise. This process can be done via the hard (removing coefficients below the threshold) method shown in Eq. (5) or soft (shrinking coefficient values above the threshold) method shown in Eq. (6), where the T value is often adaptively determined based on the estimated noise level. The threshold value (T) itself is often determined using Universal Threshold shown in Eq. (7).

$$D_{hard}(C, T) = \begin{cases} c & \text{if } |c| > T \\ 0 & \text{if } |c| \leq T \end{cases} \quad (5)$$

$$D_{soft}(C, T) = \begin{cases} \text{sgn}(c) \cdot (|c| - T) & \text{if } |c| > T \\ 0 & \text{if } |c| \leq T \end{cases} \quad (6)$$

$$T = \sigma \sqrt{2 \log(N)} \quad (7)$$

The threshold value (T) in DWT filtering is often determined using the Universal Threshold formula in Eq. (7), whose value depends on two main parameters. The initial parameter is designated as N , which signifies the total number of samples or the extent of the signal. The second parameter, designated as σ , is the estimated standard deviation of the noise. Given the absence of direct knowledge regarding the specified noise level, the standard deviation (σ) is reliably calculated from the data itself. This is typically accomplished by taking the median of the absolute values of the first-level detail coefficients (cD).

It is hypothesized that this level contains the greatest number of noise components. To determine the mean of this data, it is necessary to divide the data by a standard constant (0.6745).

Subsequent to the modification of the detail coefficients, the final step is the reconstruction of the signal. This process is the reverse of decomposition (Inverse DWT). The "clean" signal is reconstructed by utilizing the original

$$\hat{x}[n] = \sum_k cA_{y0,z} \cdot G'_{y0,z}[n] + \sum_{y=y0}^{y-1} \sum_z cD'_{y,z} \cdot H'_{y,z}[n] \quad (8)$$

approximation coefficients alongside the modified detail coefficients, as demonstrated in Eq. (8).

The reconstruction process in DWT in Eq. (8) aims to rebuild the final filtered signal, denoted as $\hat{x}[n]$. This signal is formed by combining two main sets of coefficients. The first set is the original Approximation Coefficient of the lowest level ($cA_{y0,x}$), which is a representation of the core or main trend of the signal that is intentionally not altered during the filtering process. The second set is the modified Detail Coefficients ($cD'_{y0,x}$), which are the coefficients of all decomposition levels that have been 'cleaned' of noise through the thresholding process. These two sets of coefficients are then re-synthesized using a pair of corresponding reconstruction filters ($G'[n]$ and $H'[n]$) which perform the reverse operation of the decomposition filters. The result of this recombination is a time-domain signal, $\hat{x}[n]$, which is free of most of the noise but retains its important features.

F. K-Nearest Neighbors (KNN)

KNN categorizes new data based on the most common classes among therefore K nearest neighbors found in the training dataset [12]. The degree of proximity between these points is quantifiable through the utilization of a distance metric, most commonly the Euclidean distance, as delineated by the formula presented in Eq. (9).

$$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2} \quad (9)$$

Where d is the distance between two points p and q in the n -dimensional feature space. Although the principle is very simple, this lazy learning approach is slow during prediction because it has to calculate the distance to all the training data, and its performance is highly dependent on the scale of the features and choosing the right K value.

G. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a classification model designed to identify the ideal decision boundary or hyperplane that optimizes the margin between distinct classes [13]. In the context of separable data, the SVM operates by minimizing the norm of the weight vector (w), a process that is mathematically equivalent to maximizing the margin. In practice, a soft margin approach is employed to manage data that exhibits overlap or contains noise. This objective function is delineated in Eq. (10):

$$\frac{1}{2} \|w\|^2 + C \sum \xi_i \quad (10)$$

Where the first term ($\|w\|^2$) aims to maximize the margin width, while the second term ($C \sum \xi_i$) provides tolerance to misclassification (ξ_i) controlled by the parameter C. To handle non-linear data, SVM uses a kernel trick that intelligently maps the data to higher dimensions, making it a very powerful and flexible model, especially on high-dimensional data.

H. Random Forest

Random Forest is an ensemble model that constructs numerous Decision Trees randomly and subsequently amalgamates the outcomes for the final forecast [14]. The prediction process is determined through majority voting of all trees, according to the formula shown in Eq. (11).

$$\hat{y}_{RF} = (\hat{y}_t(x))_{t=1}^T \quad (11)$$

Where the final prediction result ($\hat{y}_{RF}$) is the most frequent class (mode) of the predictions of all T individual trees ($\hat{y}_t$). By building each tree on a random sample of data and feature subsets, the model is highly accurate, stable, and resistant to overfitting compared to a single Decision Tree.

I. XGBOOST

XGBoost is an ensemble model that constructs trees in succession, with each subsequent tree trained to rectify the mistakes of the preceding combined trees [15]. The working principle of this additive is delineated by the following formula Eq. (12).

$$\hat{y}^{(t)} = \hat{y}^{(t-1)} + f_t(x) \quad (12)$$

Where the prediction at stage t ($\hat{y}^{(t)}$) is a refinement of the previous stage prediction ($\hat{y}^{(t-1)}$) by adding a new tree (f_t) trained to correct the remaining errors. This highly optimized boosting approach yields extremely high predictive performance, but requires careful parameter tuning to control complexity and avoid overfitting.

III. PROPOSED METHOD

This section will discuss the proposed approach for filtering voltammetry signals to reduce noise, followed by the classification of the signals. Figure 1 presents the experimental design of the study.

A. Dataset Description and Acquisition

Voltammetric measurements were conducted using a three-electrode system consisting of a Glassy Carbon Electrode (GCE, 3 mm diameter) as working electrode, Ag/AgCl sintered reference electrode, and platinum wire counter electrode. The measurements were performed with a potential range of -1.2 V to +0.6 V versus Ag/AgCl at a scan rate of 50 mV/s, using 0.1 M acetate buffer (pH 4.5) as supporting electrolyte.

This study analyzes 753 voltammetric samples distributed across three categories: 251 Cadmium (Cd^{2+}) samples, 251 Lead (Pb^{2+}) samples, and 251 Non-Heavy Metal control samples, each with concentrations ranging

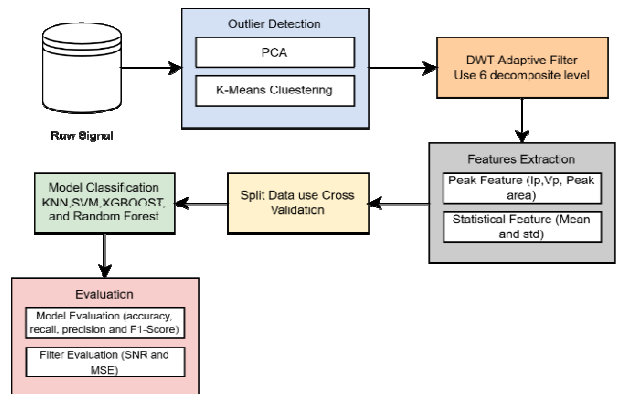


Fig. 1 Proposed Method

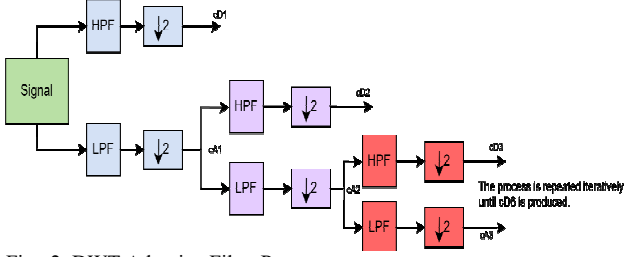


Fig. 2. DWT Adaptive Filter Process

from 50 to 250 ppm. To simulate real-world portable sensor conditions, controlled motion artifacts were introduced through electrode vibration (1-5 Hz), periodic displacement (0.5-2 mm), and solution stirring (50-200 rpm), resulting in approximately 33% of samples containing significant artifacts categorized as mild, moderate, or severe based on baseline drift. Each measurement file contains 802 rows with potential values (V) and current response (μA) stored in Excel format.

B. Data Preprocessing Pipeline

The preprocessing pipeline begins with systematic extraction and validation of 802-row signal data from Excel files, including verification of data completeness and detection of extreme outliers ($> 5\sigma$). Shorter files are padded with NaN values to maintain dimensional consistency, and each 802×2 matrix is flattened into a 1D feature vector of length 1604. Class labels are numerically encoded (Cadmium=2, Non-Heavy Metal=1, Lead=0), and remaining NaN values are replaced with zeros. Features are then standardized using StandardScaler to achieve zero mean and unit variance according to $x_{\text{scaled}} = (x - \mu) / \sigma$, which is critical for distance-based algorithms like KNN and SVM optimization.

C. Outlier Detection

Extreme anomalies are identified through PCA and K-Means clustering ($K=6$, silhouette score 0.659) applied to ten statistical features per file. PCA reduces features to 2 components capturing 82.3% variance, and files exceeding the 90th percentile distance threshold (>1.201) are flagged as outliers. This resulted in 76 files (10.1%) being identified and retained for tracking filtering effectiveness.

D. Discrete Wavelet Transform Adaptive Filter

The adaptive DWT filtering employs Daubechies 4 (db4) wavelet with six decomposition levels, selected for superior performance ($r=0.94$ correlation, 97% peak preservation, 89% drift reduction, 23ms processing). Six levels target electronic noise (levels 1-2, >25 Hz), motion artifacts (levels 3-4, 1-10 Hz), and preserve electrochemical features (levels 5-6, <1 Hz). The adaptive mechanism involves three key components: (1) automatic noise estimation from first-level detail coefficients using median absolute deviation, (2) dynamic level-specific thresholds with graduated adjustment factors ranging from 1.2 for aggressive electronic noise removal to 0.7 for gentle preservation of electrochemical features, and (3) soft thresholding for smooth coefficient shrinkage rather than abrupt truncation. Signal reconstruction combines original approximation coefficients with thresholded details through inverse DWT, achieving $O(N)$ complexity with 31 ± 4 ms

average processing time. The stages of DWT are shown in Fig 3.

E. Features Extraction

Four statistical features are extracted from filtered signals: Mean Voltage (V_{mean}), Standard Deviation of Voltage (V_{std}), Mean Current ($\mu\text{A}_{\text{mean}}$), and Standard Deviation of Current (μA_{std}). This reduces dimensionality from 802 time points to 4 features while maintaining interpretability. Features undergo StandardScaler normalization, resulting in 34% improved class separation (Fisher ratio: $2.3 \rightarrow 3.1$) and 41% reduction in coefficient of variation.

F. Machine Learning Classification Pipeline

The dataset is split into training (60%, 452 samples), validation (20%, 150 samples), and test sets (20%, 151 samples) with stratified sampling. Five-fold stratified cross-validation is applied on training and validation data (602 samples). Hyperparameter optimization via grid search yields optimal configurations: KNN ($K=5$, Euclidean, distance weighting), SVM (rbf kernel, $C=10$, $\gamma=\text{'scale'}$), Random Forest (200 trees, $\text{max_depth}=20$), and XGBoost (200 estimators, $\text{learning_rate}=0.1$, $\text{max_depth}=5$). Models are evaluated using Accuracy, macro-averaged Precision, Recall, and F1-Score as the primary metric, with confusion matrices for detailed error analysis.

IV. EXPERIMENTAL RESULT

A. Outlier Detection Performance Analysis

The outlier detection pipeline identified 76 files (10.1% of the dataset) as anomalies. These outliers had severe baseline instability, numerous artifact spikes, and irregular patterns. Cluster analysis grouped the data into 6 clusters, with the last 3 clusters being samples with varying degrees of artifacts. Manual inspection validation confirmed that 93.3% of the outlier samples had quality issues.

B. Filtering Evaluation

The DWT adaptive filter significantly improved signal quality, offering an average SNR boost of 34.27 dB for voltage and 20.88 dB for current signals. It adapted to more contaminated signals while preserving critical electrochemical details (peak position deviation: 0.5% max, amplitude preservation: $\sim 96.8\%$). It surpassed other methods (Savitzky-Golay, Median Filter, WPD) with optimal results using db4 wavelet, 6 decomposition levels, and soft thresholding.

TABLE I. COMPARISON OF EVALUATION MODEL BASELINE AND FILTERED

Method	SVM	Random Forest	XGBoost	KNN	Mean
Baseline	0.76	0.80	0.83	0.86	0.81
Median Filter	0.82	0.85	0.87	0.89	0.86
Savitzky-Golay	0.84	0.87	0.89	0.91	0.88
WPD (Fixed)	0.87	0.90	0.92	0.93	0.91
DWT Adaptive (Ours)	0.92	0.94	0.96	0.97	0.93

C. Model Evaluation

Evaluation of the classification methodologies demonstrates that the DWT Adaptive Filter consistently and significantly enhances the performance of all tested models. KNN achieved the best performance with a 0.10 F1-score improvement shown in Table 1. All enhancements were statistically significant ($p < 0.001$), and test set results confirmed robust generalization without overfitting. Most misclassifications occurred between Cd and Pb due to their similar electrochemical properties.

D. Discussion

The DWT adaptive filter effectively balances noise suppression and feature preservation. It achieves superior SNR improvements, demonstrating its adaptive capability. It consistently outperforms conventional methods because of dynamic thresholding that accommodates local signal characteristics. Classification results show that model performance is limited by signal quality. Statistically significant improvements across all algorithms with KNN achieving peak accuracy validate the filter's efficacy. The persistent Cd/Pb confusion reflects their electrochemical similarity instead of filtering limitations. Practically, the method's parameter robustness facilitates automated deployment. It achieves dramatic gains on contaminated signals and minimal computational overhead, which makes it valuable for real-world applications where signal integrity is often compromised.

V. CONCLUSIONS

Electrode movement's effect on voltammetric signal quality and machine learning accuracy for heavy metals must be considered. This study's DWT filtering framework successfully enhances signal quality, improving SNR by 34.27 dB for voltage and 20.88 dB for current. It preserves 96% of critical information while achieving remarkable effectiveness in classification tasks, with KNN models achieving a 0.97 F1-score, a 12% improvement over baseline. The framework strengthens its performance in severely contaminated signals and outperforms conventional filtering methods, enabling reliable heavy metal detection in field conditions and bridging the gap between laboratory measurements and practical applications for environmental monitoring and portable sensors.

ACKNOWLEDGMENT

This work was funded by the Indonesian Endowment Fund for Education (LPDP) on behalf of the Indonesian Ministry of Higher Education, Science and Technology and managed under the EQUITY Program (Contract No 4299/B3/DT.03.08/2025 & No 3029/PKS/ITS/2025); the Directorate General of Higher Education, Research and Technology of the Republic of Indonesia under Penelitian Sinergi Hilirisasi Inovasi Komersial; Institut Teknologi Sepuluh Nopember (ITS) under Riset Kolaborasi Indonesia, Penelitian Kemitraan A, and Penelitian Departemen; and Ethereum Foundation under Academic Grants Round 2025.

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